

Lingyun Yao

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Research Interests

- Probabilistic machine learning, tractable/neuro-symbolic models, energy-efficient AI hardware (GPU/FPGA/ASIC), approximate computing, and hardware/software co-design.

Education

- 2022 – 2026 ■ **Ph.D. in Electrical Engineering** Aalto University
Research focus: Probabilistic Computing, Neuro Symbolic AI, Energy-efficient AI.
- 2020 – 2022 ■ **M.Sc. in Micro- and Nanoelectronic Circuit Design** Aalto University
Thesis: An automated method for embedded probabilistic circuits on FPGA.
- 2016 – 2020 ■ **B.Sc. in Microelectronics Science and Engineering** Central South University

Research Publications

Journal Articles

- 1 L. Yao, S. Zhao, M. Trapp, J. Leslin, M. Verhelst, and M. Andraud, “Logsumexp: Efficient approximate logarithm acceleration for embedded tractable probabilistic reasoning,” *IEEE Transactions on Circuits and Systems I: Regular Papers*, 2025 [PDF].

Conference Proceedings

- 1 L. Yao, S. Zhao, M. Verhelst, and M. Andraud, “EinChip: A 4.6/1.5 TOPS/W 6/24b Log-compute Einsum-based accelerator for Neuro-Symbolic AI,” in *IEEE Custom Integrated Circuits Conference (CICC)*, Accepted for publication (not yet publicly available), 2026.
- 2 M. Andraud, L. Yao, K. Periasamy, J. Leslin, and M. Trapp, “Computing tractable probabilistic models: A hardware perspective,” in *Eighth Workshop on Tractable Probabilistic Modeling*, 2025 [PDF].
- 3 K. Periasamy, J. Leslin, A. Korsman, L. Yao, and M. Andraud, “Autopc: An open-source framework for efficient probabilistic reasoning on fpga hardware,” in *2024 22nd IEEE Interregional NEWCAS Conference*, 2024 [PDF].
- 4 L. Yao et al., “On hardware-efficient inference in probabilistic circuits,” in *Proceedings of the Fortieth Conference on Uncertainty in Artificial Intelligence*, 2024 [PDF].
- 5 L. Yao, M. Trapp, K. Periasamy, J. Leslin, G. Singh, and M. Andraud, “Logarithm-approximate floating-point multiplier for hardware-efficient inference in probabilistic circuits,” in *The 6th Workshop on Tractable Probabilistic Modeling*, 2023 [PDF].
- 6 J. Leslin, A. Hyttinen, K. Periasamy, L. Yao, M. Trapp, and M. Andraud, “A hardware perspective to evaluating probabilistic circuits,” in *International Conference on Probabilistic Graphical Models*, 2022 [PDF].

Under Review

- 1 S. Golipoor, L. Yao, M. Andraud, and S. Sigg, “Low-power on-device gesture recognition with einsum networks,” Submitted to *IEEE Sensors Journal*, 2026.

- 2 L. Yao et al., “Speeding up log-sum-exp: Kernel fusion at the memory wall, Integer arithmetic at the compute wall,” Submitted to NeurIPS 2026, 2026.

Selected Research Highlights

- **GPU acceleration of log-domain inference.** Designed approximate LogSumExp CUDA kernels for general-purpose GPUs, achieving **6–10× speedups** over standard PyTorch kernels.
- **Neuro-symbolic AI accelerator (16 nm ASIC tape-out).** Designed and silicon-tested a 4 mm² 16 nm chip for both neural and symbolic acceleration, reaching **8.5× higher throughput** than the symbolic baseline with near-SotA neural performance.
- **Energy-efficient probabilistic inference via approximate computing.** Built inference engines using approximate log-domain arithmetic with software-side error correction, delivering up to **649× energy reduction** and **46% area / 32% power** versus floating-point baselines, with negligible numerical error.
- **Hardware/software co-design toolflow and benchmarking.** Co-developed an open-source toolchain that maps trained probabilistic models to FPGA, and ran systematic FPGA/GPU benchmarking to derive design guidelines for future probabilistic AI accelerators.

Internship / Research Visits

- 2021-04 – 2022-10 ■ **Research Assistant, Aalto:** FPGA implementation of AI models; performance analysis; automated SW/HW flow.
- 2024-10 – 2025-02 ■ **Visiting Scholar, KU Leuven:** Finalizing our AI Chip Backend for tape out.
- 2025-08 – 2025-09 ■ **Visiting Scholar, KU Leuven:** Chip testing, and performance analysis.

Teaching & Supervising (Aalto University)

- 2026 ■ **B.Sc. thesis advisor**, approximate log-domain arithmetic for AI accelerators.
- 2024 ■ **M.Sc. thesis co-advisor**, residue number systems for hardware probability models.
- 2023–25 ■ **IC Design Project supervisor**, FP, Posit, LNS MAC unit design
- 2024, 2026 ■ **Course assistant**, Electronic Circuit, Integrated Analog Systems D.

Academic Activities

- 2024-06 ■ **Presenter**, Nordic Probabilistic AI Summer School, Copenhagen, Denmark.
- 2025-07 ■ **Co-organizer**, 8th Workshop on Tractable Probabilistic Modeling (TPM 2025), co-located with UAI 2025. *From Logic to Probabilities and Back*.
- 2026-03 ■ **Participant**, Dagstuhl Seminar: Tensor Factorizations Meet Probabilistic Circuits, Germany.

Awards & Grants

- **HPY Research Foundation Grant:** *AI-based Self-Adaptive RF Transceiver Using Hardware-Efficient Probabilistic Models*.
- **Aalto University:** Full Admission Scholarship for Master program; Dean Scholarship; Incentive Scholarship.
- **Central South University Scholarships:** Second Academic Prize (2018–2019); Third Academic Prize (2016–2017, 2017–2018); National Encouragement Scholarship of China.